Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • Grade: 100%

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • Grade: 85.71%

Lesson 5: Orthogonal Complement and Projection

- **Introduction to Orthogonal Complement**
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min
- Importance of Projections
  Reading • 4 min
- Projection Onto One-Dimensional Subspaces (Lines)
  Reading • 15 min
- Projection Onto General Subspaces
  Reading • 10 min
- Projections Case Study
  Reading • 10 min

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Introduction to Orthogonal Complement

0:07 / 1:15 2x

Read Section 3.6: “Orthogonal Complement” from Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong, and read Orthogonal Complement on Wolfram MathWorld.

Key Concepts:

- **Orthogonal Complement:** Learn what an orthogonal complement is and how it relates to subspaces in vector spaces.
- **Properties and Calculation:** Understand the properties of the orthogonal complement and how to calculate it for given subspaces.
- **Applications in Machine Learning:** Explore the significance of the orthogonal complement in various machine learning algorithms, particularly dimensionality reduction and optimization.

As you read, consider the following questions:

1. What is the orthogonal complement of a subspace, and why is it important in linear algebra?
2. How can you determine the orthogonal complement of a given subspace?
3. What are the key properties of the orthogonal complement, and how do they facilitate vector space decomposition?
4. How does the concept of the orthogonal complement apply to practical problems in machine learning, such as dimensionality reduction and optimization?
5. Why is understanding orthogonal complements crucial for grasping more advanced topics in linear algebra and machine learning?

Go to next item Completed

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